

Critical Summary: Decentralized Detection with Many Sensors

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1 Introduction

1.1 Background

Hypothesis Testing problems address the problem of deciding, based on observations, which of two or more probability laws have generated the observed data.

1.2 Decentralized Remote Hypothesis Testing

In our work, we are going to discuss the results of the paper "Decentralized Detection with Many Sensors: Optimality of Exchangeable and Identical Encoding Policies" [1]. The paper addresses a problem of Decentralized Hypothesis Testing - many sensors observe an environment, compress the observation into an action, and send the actions to a fusion center (FC). The FC decides between two hypotheses - H_1 or H_2 .

2 Problem Setup

Assuming there are two hypotheses, H_1 and H_2 , the true one can be regarded as H^* , a random variable with $P(H^* = H_1) = p$, $P(H^* = H_2) = 1 - p$. The system setup is:

- $N < \infty$ sensors, each makes an observation of the environment denoted by $y^i \in \mathcal{Y}^i$ as the observation of the i -th sensor for $i \in \{1, \dots, N\}$.
- Each sensor quantizes its observation using a mapping $\gamma^i : \mathcal{Y}^i \rightarrow \mathcal{U}^i$, $\mathcal{U}^i$ being the set of actions the sensor can send. γ^i is the decision rule / policy of the i -th sensor.
- The FC's policy is noted as $\gamma^0 : \mathcal{U}^0 \times \mathcal{U}^1 \times \dots \mathcal{U}^N \rightarrow \{1, 2\}$.
Essentially, the fusion center decides $u_0 = \gamma^0(u^{1:N})$, $u_0 \in \{1, 2\}$.
Denote the set Γ^i as the set of policies of sensor i , Γ^0 as the set of policies for the FC.
- The cost function we will focus on is:

$$c(H_j, u^0) = \begin{cases} 1 & \text{if } u^0 \neq j \\ 0 & \text{if } u^0 = j \end{cases} \quad (1)$$

- Our goal is to minimize the expected cost:

$$J^N(\gamma^{0:N}) \triangleq \mathbb{E}[c(H, u^0)] \quad (2)$$

2.1 Assumptions

In order to make the problem tractable, the authors of the paper consider a few assumptions:

- (i) $y^1, \dots, y^N$ are i.i.d. conditioned on H
- (ii) Under either hypothesis, sensor i 's observation has a probability density, measured against one common reference measure shared across both hypotheses.
- (iii) $\mathcal{U}^i = \mathcal{U}$ for $i = 1, \dots, N$, and $\mathcal{U}$ is a finite set.
- (iv) The likelihood ratio:

$$l^i = L(y^i) := \frac{f_{y^i|H}(y^i|H_2)}{f_{y^i|H}(y^i|H_1)} \text{ a.s.} \quad (3)$$

is well-behaved and invertible.

- (v) Define $I_N^{KCR} \triangleq \{u^{1:N}, \gamma^{1:N}\}$ to be the Known Common Randomization set, that includes all actions and policies of the sensors. We assume this set is known to the FC.

While (i) is a common assumption in order to simplify the math and making it tractable, assumptions (ii) and (iv) are essential in order for l^i to be a sufficient statistic of y^i , as will be discussed later. Assumption (iii) dictates that all sensors have the same action set, underlying assumption about the compression and rate being used. This will be discussed furthermore in section *, where we will offer some modifications for the solution setup. Assumption (v)'s essentiality will be discussed in Section 3.3.

3 Sufficient Statistics, Existence of an Optimal Encoding Policy

3.1 Exponential Decay of the Error Probability

According to the law of large numbers, we can expect that $J^N(\gamma^{0:N})$ will converge to the probability of error under decision rules $\gamma^{0:N}$. Also, under the criterion of minimizing the probability of error, using the MAP rule at the fusion center is optimal for any sensor policy [2,8-10].

[Proposition 5.1, 2] - if $y^1, \dots, y^N$ are i.i.d. given H^* and $P_{Y|H_1} \neq P_{Y|H_2}$, then there exists a decision rule γ' and constants $\alpha, \beta > 0$ independent of N such that if all sensors use γ' and the FC uses MAP then: $P_e < \alpha e^{-N\beta}$.

Since our FC uses MAP and $y^1, \dots, y^N$ are i.i.d. conditioned on H^* , there exists a vector of policies $\gamma^{0:N'}$ such that $J^N(\gamma^{0:N'}) < \alpha e^{-N\beta}$, it is sure that there exists $\alpha, \beta > 0$ such that:

$$J^{N*}(\gamma^{0:N}) = \inf_{\gamma^{0:N}} \{J^N(\gamma^{0:N})\} \leq J^N(\gamma^{0:N'}) < \alpha e^{-N\beta} \quad (4)$$

From equation (4) we can understand that P_e of any relevant sequence of policies is exponentially decaying as N grows large, hence we need to study and optimize the **Error Exponent**, to study how fast the error will decay given a sequence of policies. This leads to the definition:

$$J_{EE}^N(\gamma^{0:N}) = \frac{\log(J^N(\gamma^{0:N}))}{N}$$

We should notice that $J^N(\gamma^{0:N}) \leq 1$ is a probability measure meaning its log is negative, hence we want to minimize $J_{EE}^N(\gamma^{0:N})$ to get a faster decay rate.

3.2 l^i is a Sufficient Statistic

l^i is a sufficient statistic of H : once its value is known, nothing else about the raw observation y^i can further move one's belief about which hypothesis is true. With only two hypotheses on the table, the entire way in which the observation's density can vary with H is exhausted by a single comparison - how large the density is under H_2 relative to how large it is under H_1 , at that particular observation. A single number is therefore inherently enough to hold everything that changes with H , and that number is, by construction, exactly l^i . (How this shows up concretely inside the paper's own machinery — via Lemma 1 — is worth treating separately once we get there.)

No information loss: a Markov-chain and DPI reading

Sufficiency means that any two raw observations producing the same l^i are, for the purpose of distinguishing H_1 from H_2 , interchangeable: whatever detail of y^i is discarded in forming l^i carries no evidential weight about H . This has a clean information-theoretic statement. Since l^i is a deterministic function of y^i , the chain

$$H \rightarrow y^i \rightarrow l^i$$

is trivially Markov, and the Data Processing Inequality (DPI) gives

$$I(H; l^i) \leq I(H; y^i). \quad (5)$$

Equality holds *iff* l^i is sufficient for H — equivalently, *iff* $H \rightarrow l^i \rightarrow y^i$ is *also* a Markov chain, i.e. y^i carries no further information about H once l^i is fixed. This gives equality:

$$I(H; l^i) = I(H; y^i). \quad (6)$$

3.3 Optimality of Threshold Policies and the MAP Decoder

Lemma 1

For any fusion center decision rule γ^0 ,

$$\inf_{\gamma^{1:N} \in \Gamma^N} J^N(\gamma^0, \gamma^{1:N}) = \inf_{\gamma^{1:N} \in \Gamma_{TS}^N} J^N(\gamma^0, \gamma^{1:N}). \quad (7)$$

where Γ_{TS}^N is the class of threshold policies. Fixing every sensor's policy except sensor i 's, sensor i 's optimization is:

$$\inf_{u^i \in \mathcal{U}} \left\{ \sum_{j=1,2} P(H_j | y^i) E_{\gamma^{-i*}} [c(H_j, \gamma^0(u^{1:N})) | H_j] \right\} \quad (8)$$

By Bayes' rule, $P(H_j | y^i)$ depends on y^i only through the likelihood ratio l^i — this is exactly the sufficiency of l^i from Section 3.2 showing up again, now inside the optimization itself. Substituting reduces the problem to

$$\inf_{u^i \in \mathcal{U}} \left[g^i(H_1, u^i) + g^i(H_2, u^i) l^i \frac{P(H_2)}{P(H_1)} \right], \quad (9)$$

where $g^i(H_j, u^i)$ is the expected cost of action u^i under H_j given the other sensors' fixed policies. This is linear in l^i for each action, so the best action is whichever line is lowest at that value of l^i , switching only at the points where lines cross — a threshold rule.

Lemma 2

Under the assumptions above, for any deterministic encoding policy of the sensors, the optimal policy of the fusion center is given by:

$$\gamma^{0*}(u^{1:N}) = \begin{cases} H_1 & \text{if } \Delta_N(u^{1:N}; \gamma^{1:N}) \geq t \\ H_2 & \text{if } \Delta_N(u^{1:N}; \gamma^{1:N}) < t \end{cases} \quad (10)$$

for some $t \in \mathbb{R}$ and $\Delta_N(u^{1:N}; \gamma^{1:N})$ given by:

$$\Delta_N(u^{1:N}; \gamma^{1:N}) \triangleq \frac{1}{N} \sum_{i=1}^N \log \frac{P^{\gamma^i}(u^i|H_1)}{P^{\gamma^i}(u^i|H_2)} \quad (11)$$

Recalling that in [2,8-10] it was established that the MAP rule is optimal for any deterministic sensor policy, we can see where $\Delta_N(u^{1:N}; \gamma^{1:N})$ is coming from. The MAP rule for our problem is:

$$\gamma^{0*}(u^{1:N}) = \arg \max_{i \in \{1,2\}} P(H_i|u^{1:N}) \quad (12)$$

It can be written in the form of a log likelihood decision rule:

$$\gamma^{0*}(u^{1:N}) = \begin{cases} H_1 & \text{if } \log \frac{P(H_1|u^{1:N})}{P(H_2|u^{1:N})} \geq 0 \\ H_2 & \text{if } \log \frac{P(H_1|u^{1:N})}{P(H_2|u^{1:N})} < 0 \end{cases} \quad (13)$$

And looking at the log likelihood:

$$\log \frac{P(H_1|u^{1:N})}{P(H_2|u^{1:N})} = \log \frac{P(u^{1:N}|H_1)}{P(u^{1:N}|H_2)} + \log \frac{P(H_1)}{P(H_2)} \quad (14)$$

$$= \log \prod_{i=1}^N \frac{P(u^i|H_1)}{P(u^i|H_2)} + \underbrace{\log \frac{p}{1-p}}_{\triangleq -Nt} \quad (15)$$

$$= \sum_{i=1}^N \log \frac{P(u^i|H_1)}{P(u^i|H_2)} - Nt \quad (16)$$

Here, (14) follows from Bayes' rule, and (15) from y^i being i.i.d. given H . We should just note that:

$$\sum_{i=1}^N \log \frac{P(u^i|H_1)}{P(u^i|H_2)} - Nt \geq 0 \iff \frac{1}{N} \sum_{i=1}^N \log \frac{P(u^i|H_1)}{P(u^i|H_2)} \geq t \quad (17)$$

And this sums up how the optimal decision rule of the FC is established. It is important to note that this decision rule is why assumption (v) is important for the paper's solution: in order to calculate $\Delta_N(u^{1:N}; \gamma^{1:N})$, the FC must know how u^i was derived, to give it its proper weight in the decision process.

3.4 Existence of an Optimal Solution

Lemmas 1 and 2 tell us where to look; Theorem 1, via the compactness result of Lemma 3, is what guarantees something is actually there to find.

Lemma 3

Γ_{TS}^N is compact under total variation.

Lemma 3 is an important step towards establishing the **existence** of an optimal solution, the paper's main contribution to the finite N problem.

Theorem 1

There exists an optimal decision rule sequence $\gamma^{0:N*} = (\gamma^{0*}, \gamma^{1*}, \dots, \gamma^{N*})$. These policies are MAP for the FC and threshold for the sensors.

The proof of Theorem 1 comes from Lemmas 1 and 2, regarding how the optimal decision rule sequences behave, and ensures the existence of such a decision rule sequence by showing that the infimum of $J^N(\gamma^{0:N})$ is continuous in the sensors' policies, and using Weierstrass's extreme value theorem, it is shown that it must attain a minimum there.